Trastuzumab scFv vs HER2: Hackathon Workflow

The task is affinity maturation of a known antibody with a very small wet-lab budget (15 variants). The whole game is to **use computation to pick 15 sequences that are each likely to beat 0.82 nM, and that don't all fail for the same reason.**

1. What to start with

- **The parent sequence (242 aa).** It's on the starting-point slide. Check it is VH, $(GGGGS)_3$, VL, with a length of 242.
- **PDB 1N8Z**, the HER2 ectodomain bound to the trastuzumab Fab. Chain B is the VH, chain A is the VL, and the HER2 chain is the antigen.
- **HER2 ECD sequence**, UniProt P04626, residues 23-652, for cofolding.
- **The rules, as a validator script.** Write this early, because it will save you at submission time:
  - At least 5 mutations from the parent
  - Single chain, in the order VH, $(GGGGS)_3$, VL
  - At most 260 aa, with only standard amino acids
- **The EGFR.md campaign prompt** from the HuggingFace dataset, as a template for how to structure the work. The ProteinBase protein design skills are worth a look if you are using Claude Code to drive the tools.

2. Phase 0: Baseline and structural map

1. From 1N8Z, compute the **paratope**: residues within about 4-5 Å of HER2, plus the CDRs. Number the antibody with ANARCI or similar, so you can talk in CDR terms.
2. **Cofold the parent with HER2 in Boltz-2** and compute its **ipSAE**. This is your baseline. Check that the prediction reproduces the 1N8Z interface. If it doesn't, ipSAE is a weak signal for you and you should lean more on the sequence models.
3. Trastuzumab binds the membrane-proximal domain (domain IV). Cropping HER2 to that region will make cofolding much faster. Check whether the organizers' ipSAE scoring expects the full ectodomain.

3. Phase 1: Generate candidate mutations (cast a wide net)

Tool	What it gives you
ESM2 / ESM3	Zero-shot single and double mutant likelihoods. Good at flagging mutations that are evolutionarily plausible, so they act as a "does this destabilize things" filter.
AbLang2	Antibody-specific masked scan across framework and CDRs. More antibody-aware than general PLMs, including paired VH/VL context.
DSM	Sequence-only diffusion edits. Good for generating diverse multi-mutation variants without structure.
ProFam-1	Family-as-prompt scoring. Its role is a second, independent opinion on fitness.
AbMPNN	Backbone-fixed redesign from 1N8Z. Your structure-aware source, mainly for interface residues.
Boltz-2	Cofolding plus the ipSAE interface score.
Mosaic	Combines several objectives (Boltz, AF2, MPNN) into one optimization.
EVOLVEpro	Few-shot active learning, but it needs labelled data.

A note on EVOLVEpro: you only get one wet-lab round, with results about 3 weeks later. So there is no real labelled loop during the hackathon. It is only useful if you use a proxy score as an in-silico oracle, or find public trastuzumab-HER2 binding data to train on. Treat it as optional.

4. Phase 2: Score and filter

Build a table with one row per candidate and these columns:

- Mutation list and count
- PLM scores (ESM, AbLang2, ProFam)
- Boltz-2 ipTM, ipSAE, pLDDT, and the change in ipSAE versus the parent

- Developability flags, computable with simple scripts. The main ones are new N-glycosylation motifs (N-X-S/T), unpaired cysteines, and big new hydrophobic patches. For an scFv that has to express well, also be conservative with framework and VH/VL interface mutations.

Rank by consensus rather than any single score. A mutation that several independent methods like is much safer than one a single model loves.

5. Phase 3: Design the pool of 15

You are judged on the pool, so think of it as a portfolio rather than 15 copies of your best guess:

- **Conservative variants:** the highest-consensus mutations, combined to reach at least 5.
- **Moderate variants:** mixes of CDR and interface mutations from different sources.
- **Bolder variants:** more novel or structure-driven designs, kept in a smaller share.

Avoid having every variant share the same risky mutation. Also check what the ipSAE of the pool means in the rules. Whether it is per-variant averaged or something else changes how you should diversify, so ask the organizers if it isn't clear.

6. Where you end up

- A **FASTA/CSV of 15 sequences** that pass your validator, with the mutation list for each versus the parent
- A **score table** from your pipeline, with the pool-level ipSAE
- A **short write-up** covering which tools you used and why, and your reasoning for how you balanced the pool. Since "creative approach" and "tools you used and why" are judging criteria, keep a running log as you go.

Rough time split

Stage	Share
Setup and baseline	~20%
Candidate generation	~30%
Scoring and filtering	~25%
Pool design and validation	~15%

Stage	Share
Write-up	~10%

Caveat: some of these tools (ProFam-1, Mosaic, DSM) are newer, and exact install steps and APIs should be checked against the hackathon's setup docs, including which tools are preinstalled and what GPU access you have. Boltz-2 in particular needs a GPU.